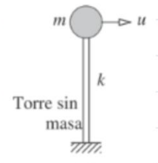
$$m\ddot{u} + ku = 0 \qquad \omega_n = \sqrt{\frac{K}{m}}$$



(b)

$$\ddot{u} = \frac{d^2u}{dt} \qquad \div m$$

$$\ddot{u} + \frac{Ku}{m} = \frac{0}{m} \qquad \omega_n^2 = \frac{K}{m}$$

$$\ddot{u} + \omega_n^2 u = 0$$

$$\ddot{u} = m^2 \qquad m^2 + \omega_n^2 = 0 \qquad m^2 = -\omega_n^2$$

$$\dot{u} = m$$

$$u = m^0 = 1 \qquad m = \pm\sqrt{-\omega_n^2}$$

$$m = \pm \omega_n i \longleftarrow m = \alpha \pm \beta i \qquad \alpha = 0$$

$$u = e^{\alpha x}(A \cos \beta t + B \operatorname{Sen} \beta t) \qquad \beta = \omega_n$$

$$u = e^{0x}(A \cos \omega_n t + B \operatorname{sen} \omega_n t)$$

$$u(t) = A \cos \omega_n t + B \operatorname{sen} \omega_n t$$


Reposo

$\dot{u}(t) = 0$

$$t = 0$$

$$u(0) = u_0$$

$$u(0) = A \cos \omega_n * 0 + B \operatorname{sen} \omega_n (0) = u_0$$

$u = 0$

$$u(0) = A + B*0 = u_0 \qquad A = u_0$$

$$\dot{u}(t) = -A\omega_n \operatorname{sen} \omega_n t + B\omega_n \cos \omega_n t$$

$$\dot{u}(t) = -u_0 \omega_n \operatorname{Sen} \omega_n t + B\omega_n \cos \omega_n t$$

$$\dot{u}(0) = -u_0 \omega_n \operatorname{Sen} \omega_n * 0 + B\omega_n \cos \omega_n * 0 = 0$$

$$\dot{u}(0) = 0 + B = 0 \qquad B = 0$$

$$u(t) = u_0 \cos \omega_n t$$

1cm

$m = 5\ \text{Kg m}$

$K = 1\ \text{N/m}$

$$\omega_n = \sqrt{\frac{K}{m}} = \sqrt{\frac{\frac{1N}{m}}{\frac{5\ N\text{-}s^2}{m}}}$$

$$1\ \text{Kg m} = \frac{1\ N\text{-}s^2}{m}$$

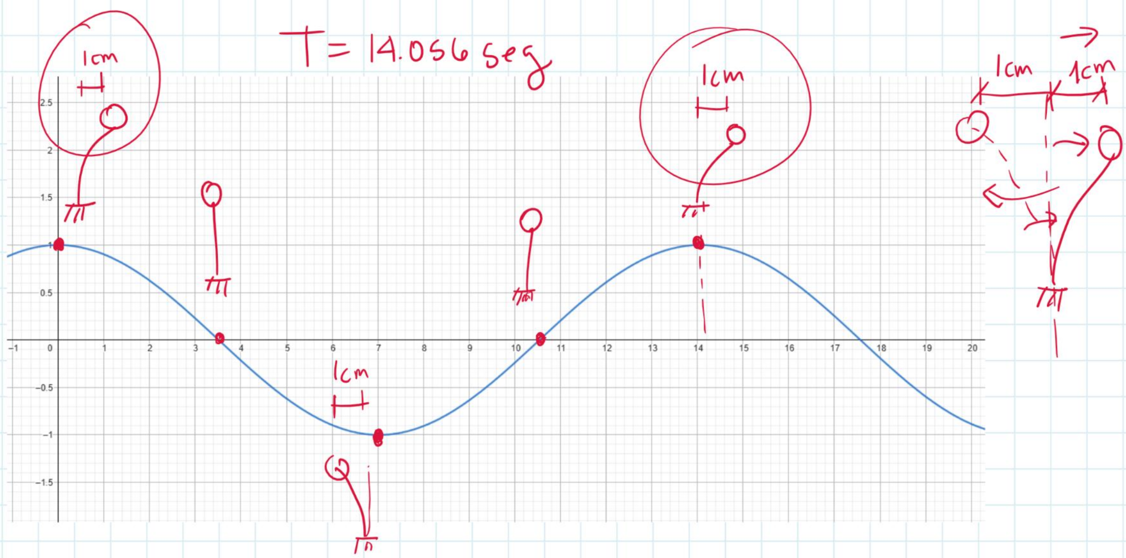
$$\omega_n = \sqrt{\frac{1N\text{-}\cancel{m}}{5N\text{-}\cancel{m}\text{-}s^2}} = \omega_n = \sqrt{\frac{1}{5\ s^2}}$$

$$\omega_n = 0.447\ s^{-1} = 0.447\ Hz$$

$$T = \frac{2\pi}{\omega_n} = \frac{2\pi}{0.447}$$

$$u(t) = 1 * \cos 0.447\, t = \cos 0.447\, t$$

$$T = 14.056\ \text{Seg}$$




a) $u(0) = 0$ $\qquad$ $\dot{u}(0) = V_0$ $\leftarrow$

b) $u(0) = u_0$ $\qquad$ $\dot{u}(0) = V_0$

$$\ddot{u} + w_n^2 u = 0$$

$u = y$ $\qquad$ $y'' + \omega^2 y = 0$

$t = x$ $\qquad$ $y(0) = 0 \qquad y'(0) = V_0$

$w_n = \omega$

$$sol := y(x) = \frac{v \sin(\omega x)}{\omega}$$

a) $u(t) = \frac{V_0}{w_n} \sin(w_n t)$

$$u(t) = A \cancel{\cos}^{\,0} w_n t + B \operatorname{sen} w_n t$$

$u(0) = 0$ $\qquad$ $A \cos w_n (0) + B \operatorname{sen} w_n (0) = 0$

$A \cos(0) + B \operatorname{sen}(0) = 0$

$A * 1 + B * 0 = 0$ $\qquad$ $A = 0$

$A + 0 = 0$

$\dot{u}(0) = V_0$

$\dot{u}(t) = 0 * \cancel{\cos}^{\,0} w_n t + B w_n * \cos w_n t$

$\dot{u}(t) = B w_n \cos w_n t$

$\dot{u}(0) = V_0$ $\qquad$ $\dot{u}(0) = B w_n \cos(w_n * 0) = V_0$

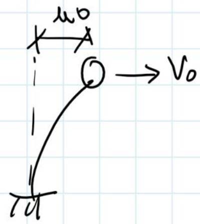
$B w_n \cos(0) = V_0$

$B w_n (1) = V_0$ $\qquad$ $B = \frac{V_0}{w_n}$

a) $u(t) = \frac{V_0}{w_n} \operatorname{sen} w_n t$

b)

$$sol := y(x) = \frac{v \sin(\omega x)}{\omega} + u \cos(\omega x)$$




$$u(t) = u_0 \cos \omega_n t + \frac{V_0}{\omega_n} \operatorname{sen} \omega_n t$$

$$u(t) = A \cos \omega_n t + B \operatorname{sen} \omega_n t$$

$u(0) = u_0 \qquad \dot{u}(0) = V_0$

$u(0) = A \cos(0) + B \operatorname{sen}(0) = u_0$

$A * 1 + B * 0 = u_0 \qquad A = u_0$

$\dot{u}(t) = u_0 (\omega_n * - \operatorname{sen} \omega_n t) + B(\omega_n \cos \omega_n t)$

$\dot{u}(t) = - u_0 \omega_n \operatorname{sen} \omega_n t + B \omega_n \cos \omega_n t$

$\dot{u}(0) = - u_0 \omega_n \overset{0}{\cancel{\operatorname{sen}}\, \omega_n * 0} + B \omega_n \cos \overset{1}{\cancel{\omega_n} * 0} = V_0$

$\dot{u}(0) = 0 + B \omega_n = V_0 \qquad B = \frac{V_0}{\omega_n}$

$$des := \frac{v \sin(\omega x)}{\omega} + u \cos(\omega x)$$

$$vel := v \cos(\omega x) - u \sin(\omega x)\, \omega$$

$$acel := -v \sin(\omega x)\, \omega - u \cos(\omega x)\, \omega^2$$


$\rightarrow u(t) = \frac{V_0}{\omega_n} \sin(\omega_n t) + u_0 \cos(\omega_n t)$ Desplazamiento

$\rightarrow \dot{u}(t) = V_0 \cos(\omega_n t) - \overset{V}{u_0 \omega_n} \sin(\omega_n t)$ Velocidad

$\rightarrow \ddot{u}(t) = -V_0 \omega_n \sin(\omega_n t) - \underset{a}{u_0 \omega_n^2} \cos(\omega_n t)$ aceleración

física

$V = \frac{u}{t} \qquad u = Vt$

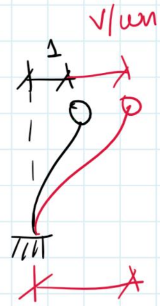
$\frac{V_0}{\omega_n} \rightarrow \frac{V}{\omega_n} = u \qquad \omega_n = \sqrt{\frac{K}{m}} \quad \left(\frac{Hz}{s^{-1}}\right)$

$V = u\, \omega_n \qquad cm * s^{-1} = \frac{cm}{s}$

$V_0 \omega_n = a \qquad a = V \omega_n \qquad a = (u \omega_n) \omega_n = u \omega_n^2$

$a = u \omega_n^2$

$\ddot{u}(0) = -V_0 \omega_n \sin(\omega_n * 0) - u_0 \omega_n^2 \cos(\omega_n 0) = - u_0 \omega_n^2$

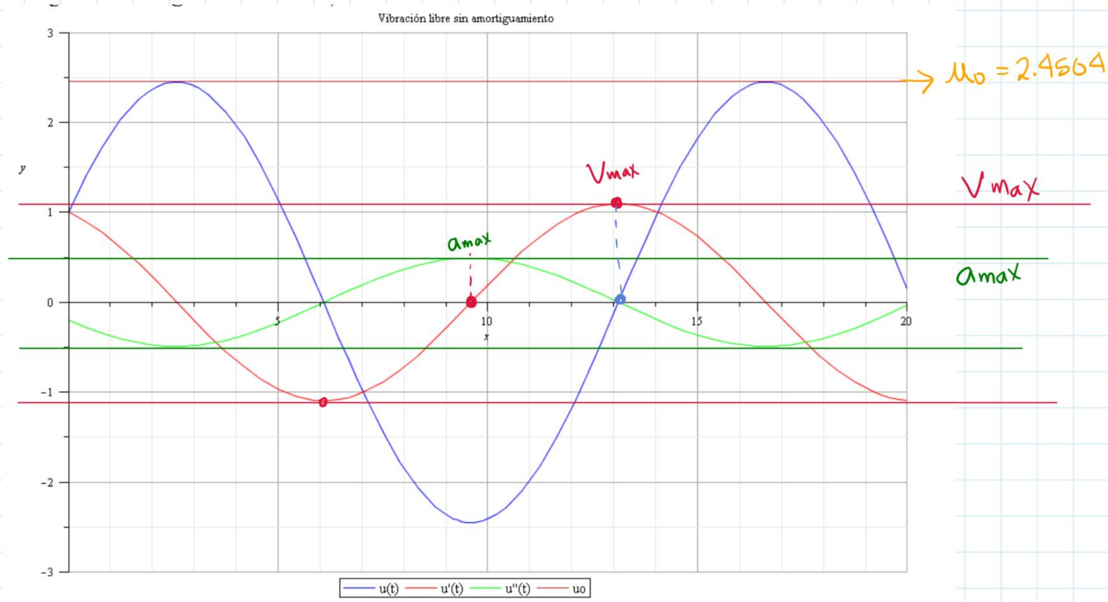



$u_0 = 1$ cm

$V_0 = 1$ cm/s

$$u_0 = \sqrt{(u_0)^2 + \left(\frac{V_0}{w_n}\right)^2}$$

$$u_0 = \sqrt{1^2 + \left(\frac{1}{0.447}\right)^2} = 2.4504\text{cm}$$




$u_{max} = 2.4504$ cm

$V_{max} = u_{max} * w_n = 2.4504 * 0.447 = 1.095$ cm/s

$V_0 = 2$ cm/s

$$u_{max} = \sqrt{1^2 + \left(\frac{2}{0.447}\right)^2} = 4.585$$

$V_{max} = 4.585 * 0.447 = 2.05$ cm/s

$a_{max} = u_{max} * w_n^2 = 4.585 * 0.447^2 = 0.916$ cm/s$^2$